

UNDERSTANDING OSI MODELS

What is the OSI model?

- Open Systems Interconnection Reference Model

It is a guide (hence the term "model")

- Avoid focusing excessively on minute details

This is distinct from the OSI protocol suite

- The majority of OSI protocols were not widely adopted

Unique protocols exist at each layer

This model serves as a constant reference throughout a networking career

- It will be referenced frequently

Mnemonic: All People Seem To Need Data Processing

Please - Physical

Do - Data Link

Not - Network

Take - Transport

Susan's - Session

Panties - Presentation

Away - Application

Layer 7 - Application

- The layer we see
- HTTPS, FTP, DNS, POP3
- "Your Eyes"

Layer 6 - Presentation

- Character Encoding
- Application Encryption
- "Application Encryption (SSL/TLS)"

Layer 5 - Session

- Communication management between devices
- Start, stop, restart

- "Control Protocol, Tunneling Protocols"

Layer 4 - Transport

- The "post office" layer
- Parcels and letters
- TCP (Transmission Control Protocol) and UDP (User Datagram Protocol)
- "TCP Segment, UDP Datagram"

Layer 3 - Network

- The "routing" layer
- Internet Protocol (IP)
- Fragments frames to traverse different networks
- "IP Address, Router, Packet"

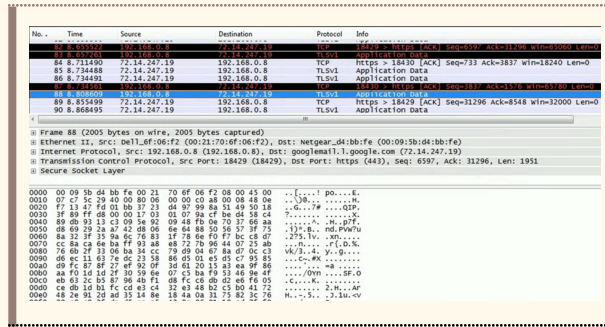
Layer 2 - Data Link

- The basic network "language"
- The foundation of communication at the data link layer
- Data Link Control (DLC) protocols
- MAC (Media Access Control) address on Ethernet
- "Frame, MAC Address, EUI - 48, EUI - 64, Switch"

Layer 1 - Physical

- The physics of the network
- Signaling, cabling, connectors
- This layer isn't about protocols
- "You have a physical layer problem."
- Fix your cabling, punch-downs, etc.
- Run loopback tests, test/replace cables, swap adapter cards
- "Cable, Fibers, and the signal itself"

Wireshark



Summary Protocol

Layer 88

- ★ Application:
https://mail.google.com
- ★ Presentation: SSL encryption
- ★ Session: Link the presentation to the transport
- ★ Transport: TCP encapsulation
- ★ Network: IP encapsulation
- ★ Data Link: Ethernet
- ★ Physical: Electrical signals

Lesson Learned:

In this lesson, I learned about the OSI Model. These are the layers used to transmit data between devices. I have already memorized the layers because they are important for understanding how data is transmitted and which protocols are used at each layer.

NETWORKING DEVICES

Many different ways to forward traffic

- A data center full of equipment
- Every device have a purpose
- The implementation may change over time
- Once installed, It can often be difficult to remove

There are new technologies all the time

- Always something to learn

Router

Routes traffic between IP subnets

- OSI layer 3 device
- Routers inside of switches sometimes called "layer 3 switches"
- Layer 2 = Switch
- Layer 3 = Router

Often connects diverse network types

- LAN, WAN, copper, fiber

Switch

Bridging done in hardware

- Application-specific integrated circuit (ASIC)

An OSI layer 2 device

- Forwards traffic based on data link address

Many ports and features

- The core of an enterprise network
- May provide Power over Ethernet (PoE)

Multilayer Switch

- Includes Layer 3 (routing) functionality

Firewalls

Filter traffic by port number or application

- Traditional vs. NGFW

Encrypt traffic

- VPN between sites

Most firewalls can be layer 3 devices (routers)

- Often sits on the ingress/egress of the network
- Network Address Translation (NAT)
- Dynamic routing

IDS and IPS

Intrusion Detection System /

Intrusion Prevention System

- Watch network traffic

Intrusions

- Exploits against operating systems, applications, etc.
- Buffer overflows, cross-site scripting, other vulnerabilities

Detection vs. Prevention

- Detection – Alarm or alert
- Prevention – Stop it before it gets into the network

Balancing the load

Distribute the load

- Multiple servers
- Invisible to the end-user

Large-scale implementations

- Web server farms, database farms

Fault tolerance

- Server outages have no effect
- Very fast convergence

Load Balancer

Configurable load

- Manage across servers

TCP offload

- Protocol overhead

SSL offload

- Encryption/Decryption

Caching

- Fast response

Prioritization

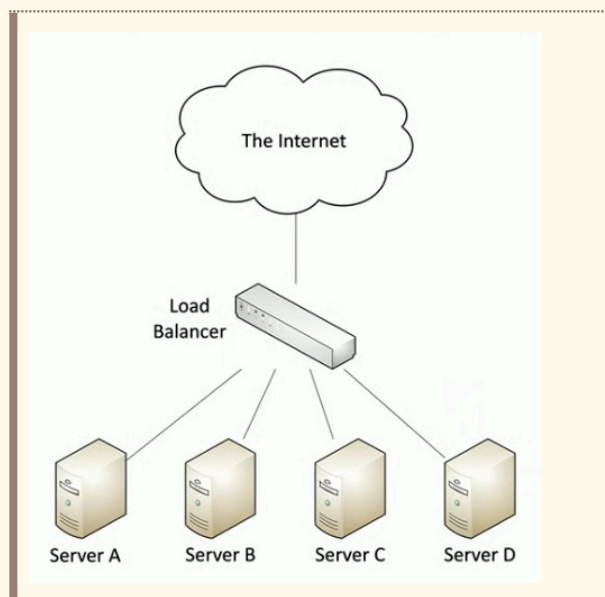
- QoS

Content switching

- Application-centric balancing

NAS vs SAN

- Network Attached Storage (NAS)
- Connect to a shared storage device across the network



Proxies

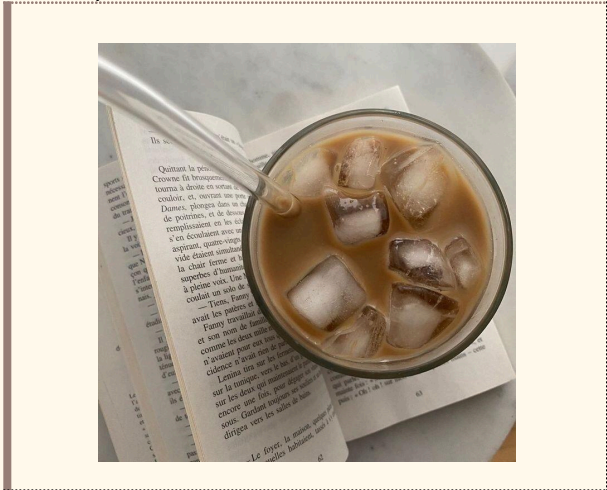
- Sits

Term	Definition
Term	Definition

Sub-Topic

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(For photos)



TERMINOLOGIES